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When AI is fairer than humans: The role of egocentrism in moral and fairness judgments of AI and human decisions

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ABSTRACT

Algorithmic fairness is a core principle of trustworthy Artificial Intelligence (AI), yet how people perceive fairness in AI decision-making remains understudied. Prior research suggests that moral and fairness judgments are egocentrically biased, favoring self-interested outcomes. Drawing on the Computers Are Social Actors (CASA) framework and egocentric ethics theory we examine whether this bias extends to AI decision-makers, comparing fairness and morality perceptions of AI and human agents. Across three experiments (two preregistered, N = 1880, Prolific US samples), participants evaluated financial decisions made by AI or human agents. Self-interest was manipulated by assigning participants to conditions where they either benefited from, were harmed by, or remained neutral to the decision outcome. Results showed that self-interest significantly biased fairness judgments—decision-makers who made unfair but personally beneficial decisions were perceived as more moral and fairer than those whose decisions benefited others (Studies 1 & 2) or those who made fair but personally costly decisions (Study 3). However, this egocentric bias was weaker for AI than for humans, mediated by a lower perceived mind and reduced liking for AI (Studies 2 & 3). These findings suggest that fairness judgments of AI are not immune to egocentric biases, but are moderated by cognitive and social perceptions of AI versus humans. Our studies challenge the assumption that algorithmic fairness alone is sufficient for achieving fair outcomes. This provides novel insight for AI deployment in high-stakes decision-making domains, highlighting the need to consider both algorithmic fairness and human biases when evaluating AI decisions.

1. When AI is fairer than humans: the role of egocentrism in moral and fairness judgments of AI and human decisions

The rapid integration of artificial intelligence (AI) into decision-making processes has raised critical questions about fairness, morality, and public trust in algorithmic systems. AI is increasingly used in high-stakes contexts such as hiring (Li et al., 2021), medical diagnostics (McKinney et al., 2020), and criminal justice (National Institute of Justice, 2018). While past research suggests that AI decision-making is often perceived as more objective than human decision-making (Helberger et al., 2020), it is also subject to unique biases in moral evaluations (Bigman & Gray, 2018). Understanding how individuals judge AI decision-makers, particularly when fairness and morality are at stake, is essential for informing AI design and public policy. As AI systems continue to influence outcomes in domains with direct human impact, insight into public moral evaluations becomes critical for ensuring that these systems align with societal values and are trusted by users.

A well-established bias in moral cognition refers to the theory of egocentric ethics—the tendency to evaluate moral and fairness judgments through the lens of self-interest (Epley & Caruso, 2004). Several studies have empirically confirmed egocentric ethics, showing that moral judgments are prone to the self-interest bias. For example, Bocian and Wojciszke (2014a) showed that judgments of different transgressions depend on the involvement of observers' interests, as they were judged more leniently when they could benefit from them and more harshly when they could not. Similarly, moral perception of in-group and out-group members depends on whether their transgressions serve interests of the group (Bocian et al., 2021).

While this self-interest bias has been robustly demonstrated in human social interactions (see Bocian et al., 2020), it remains unclear whether and how it extends to AI decision-makers. If people apply the same fairness and morality standards to AI as they do to humans, we would expect self-interest bias to emerge similarly. However, if AI is perceived as a fundamentally different type of decision-maker—one that lacks intentionality, emotions, or a "moral mind"—self-interest may

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play a reduced role in evaluating AI decisions (Gray et al., 2012).

By answering these questions, our study contributes to both the moral psychology of decision-making and the emerging field of AI ethics and fairness perception. Unlike human decision-makers, AI lacks an inherent moral compass or ethical responsibility, yet its decisions shape real-world outcomes in ways that mimic human moral agency (Bigman & Gray, 2018). Understanding how fairness perceptions of AI differ from those of humans is critical, as it informs how people trust, accept, or reject algorithmic decision-making in key societal domains (Helberger et al., 2020; Rahwan et al., 2019).

1.1. Egocentric morality in fairness and moral judgments of AI and humans

According to the theory of egocentric ethics, moral cognition is predominantly biased by the egocentric perspective, which is readily and automatically available (Epley & Caruso, 2004). Further, we typically view our judgments as objective because we are unaware of the underlying biases influencing our moral perceptions (Goodwin & Darley, 2008; Wojciszke & Bocian, 2018). Research has shown that various facets of the egocentric perspective—such as interpersonal attitudes, group interest, and self-interest—influence moral judgments, often making them more lenient (Bocian et al., 2020). For example, in the first article depicting the self-interest bias, Bocian and Wojciszke (2014a) showed that people judged a person who cheated to win an eye-catching prize as more likable and moral if their own prize was linked to the cheaters, compared to when the observer's interest was uninvolved with the cheater (Study 2; Bocian & Wojciszke, 2014a). The following studies showed that people are unaware of the self-interest bias and trust those who cheat others, since they benefited from their transgressions (Bocian et al., 2016; Bocian & Wojciszke, 2014b).

The self-interest bias is explained by increased liking toward the person deciding in our favor, which explains the differences observed in moral evaluations (Bocian & Wojciszke, 2014a). Previous studies directly show that liking impacts moral judgments; when we like someone more, we perceive them as more moral (Bocian et al., 2018). Further, the self-interest bias does not emerge if the target is disliked by the recipient, proving that liking is a mediator between self-interest and moral cognition (Bocian & Wojciszke, 2014a). The influence of self-interest is significant enough to be observed in various aspects of social interactions. Studies have found its effects in various contexts, e.g., on preschool children's attitudes towards individuals with antisocial behavior (Myslinska-Szarek et al., 2021), hypocrites (Bocian et al., 2024), intergroup conflicts (Bocian et al., 2021), judgments about laws and regulations (Bialobrzeska et al., 2015), and AI moral responsibility (Dong & Bocian, 2024).

To investigate whether the same bias will occur when AI is the decision-maker, we build on the Computers Are Social Actors (CASA) framework (Nass & Moon, 2000), which posits that people often treat AI as social agents and apply human-like expectations to their behavior. Indeed, multiple studies so far have shown that artificial agents are considered social agents (Groom & Nass, 2007; Lebrun et al., 2024). For example, Lebrun et al. (2024) showed that people experience phantom costs with both humans and robots making too generous offers. Therefore, fairness and moral judgments of AI decisions may also be prone to self-interest bias, which highlights the potential that the morality of these decisions could be judged based on personal gain rather than objective reasoning. More specifically, regarding perceptions of fairness in algorithm decision-making, previous studies found that positive outcomes for observers improve fairness judgments of the decision (Wang et al., 2020).

Moreover, in a different study, when a hiring decision was favorable for participants, they judged the decision-maker as more just when it was a human rather than an AI. However, there were no differences regarding the unfavorable hiring outcome (Bankins et al., 2022). Similarly, research showed that people want others to use altruistic autonomous vehicles but prefer self-preserving ones for themselves (Bonnefon et al., 2016). People are also less likely to report a self-beneficial error when AI (vs. human) is responsible and feel less guilty for not disclosing it (Giroux et al., 2022). Finally, Dong and Bocian (2024) showed that when people could transfer responsibility to AI advisors to serve their interests, they judged their transgressions more leniently.

Nevertheless, the direct impact of self-interest bias on fairness and moral judgments regarding human and AI decision-makers has yet to be systematically tested. Thus, we sought to scope the strength of the self-interest bias by comparing people's fairness and moral judgments toward humans and AI, which make unfair and fair financial decisions. Moreover, we investigated potential psychological mechanisms that could explain how self-interest bias works, such as interpersonal attitudes and mind perception.

If the self-interest bias is driven in part by affective responses such as liking (Bocian et al., 2018), then moral judgments regarding AI may be less susceptible to egocentric moral distortions. In the previous studies, benefiting the observer increased the immoral agent's liking, which explained the more lenient moral judgments. However, when the liking of the immoral agent was manipulated and limited, the self-interest bias did not emerge (Bocian & Wojciszke, 2014a). Hence, we tested whether this mechanism extends to the new type of agent — AI. We hypothesized that participants would like an agent who benefits them more than an agent who does not. However, this effect would be stronger for humans than for AI agents, as liking depends on similarity between the observer and judged agent (Bocian et al., 2018). Consequently, liking would drive fairness and moral character judgments differently for human and AI agents.

Egocentric bias is theoretically central to understanding how people morally evaluate AI decision-makers. Such evaluations are not strictly objective judgments of fairness or moral correctness but are often filtered through self-interest (Epley & Caruso, 2004). While existing studies have shown the discrepancy between moral and fairness judgments of AI and humans (e.g. Bigman et al., 2023; Shank et al., 2019; Wilson et al., 2022), these differences may be further amplified when viewed through the lens of egocentrism.

This matters not just for theory, but for practice. As researchers and practitioners aim to align AI systems with human values, they assume that "moral" or "fair" behavior can be defined in a relatively stable, shared way. Yet the theory of egocentric ethics (Epley & Caruso, 2004) challenges that assumption: individuals with conflicting self-interests may interpret the same AI decision differently, thus making the alignment goal unattainable. However, interpersonal attitudes are not the only possible mechanism explaining the differences in the judgements of AI and Humans.

1.2. Moral character judgments and mind perception

According to the Mind Perception Theory, mind perception lies at the core of moral judgments (Gray et al., 2012). Specifically, moral judgments are influenced by the perceived minds of two parties—the agent and the recipient—and are sensitive to two dimensions of the mind: experience and agency. Experience represents the perceived capacity for sensation and feelings, while agency denotes the perceived capacity to form intentions and take actions. Previous research has shown that adult humans typically score high in both dimensions, whereas robots are often perceived as low in agency and experience compared to humans (Bigman & Gray, 2018).

Differences in mind perception are also observed when judging AI and humans in similar moral situations. For example, people judge AI as more objective than humans and ascribe less prejudiced motivation to algorithms, making them less morally outraged by algorithms than human discrimination (Bigman et al., 2023). In effect, as mind perceptions impact how we perceive humans and AI agents, we should also observe differences in moral judgments of their moral character. This pattern was observed in previous studies, as for example Wilson et al. (2022) found that AI is judged more leniently for the same immoral actions compared to humans. Similarly, Shank et al. (2019) found that AI was perceived to be less at fault than humans for the same wrongdoings. Importantly, people are generally averse to machines making moral decisions, partly due to the absence of a "complete mind" (Bigman & Gray, 2018). If mind perception explains our moral judgments, we hypothesized that AI would be perceived as having less mind than humans and, therefore, judged as a less moral agent.

A complementary explanation focuses on the perception of AI as bias-free. Some results show that people believe AI to be fairer than humans (Helberger et al., 2020). As a result, individuals may perceive AI as operating with a higher degree of evaluative precision. For example, AI can detect subtle behavioral signals of deception (e.g., facial and head movements) more accurately than human evaluators (Suen et al., 2024). In effect, transparently disclosing AI usage can result in users behaving more strategically (Suen & Hung, 2024), to align with the system. In sum, AI is seen as lacking in mind qualities, which may result in reduced opportunity for self-justification when outcomes are unfavorable.

While previous literature mainly focused on fairness perceptions regarding moral judgments in algorithmic decision-making (ADM; for review, see Starke et al., 2022), we focused on moral character judgments. The way people perceive someone's moral character can have a significant impact on various social aspects. It can influence how one judges someone's general character, how much trust they have in them during economic transactions, and whether they are considered eligible for different social roles (Crockett et al., 2021). Moreover, these perceptions are crucial in determining who people interact with and avoid (Abele & Wojciszke, 2014). In some extreme cases, they can even affect life-or-death decisions (Wilson & Rule, 2015). Therefore, this research may contribute to a better understanding of the social consequences of self-interest bias in the modern digital world by examining the moral character judgments of humans and AI agents.

1.3. Overview of studies

Overall, in three studies, we aimed to test three hypotheses using key constructs defined in Table 1.

First, we hypothesized that self-interest bias would be visible in the moral character and fairness judgments of both AI and human agents. Thus, in Studies 1 and 2, we convinced participants that AI or human agents made an unfair financial decision that benefited participants or other parties. Further, we asked participants to report the liking and moral character of the agent (Study 1), and mind perception (Study 2). In Study 3, we further improved the self-interest manipulation by presenting AI or human agents who made either an unfair decision beneficial for participants or a fair decision beneficial for other parties. To move beyond the limitations of vignette-based studies in moral psychology (for critiques, see Mahmud et al., 2022; Doliński, 2018; Wojciszke & Bocian, 2018), all of our studies employed behavioral experimental designs, two of them preregistered. This ensures observing the self-interest bias on consequential outcomes, rather than relying solely on participants' responses to hypothetical scenarios.

Table 1
Definitions of key constructs used in all studies.

Construct	Definition
Moral Character	The perceived trustworthiness, fairness, and ethical standing of the decision-maker, assessed through trait-based judgments (Abele et al., 2016).
Fairness Perceptions	Subjective evaluation of whether a decision is just, irrespective of its procedural correctness, measured using a single-item fairness rating (Starke et al., 2022).
Self-Interest Bias	The tendency to judge decision-makers more favorably when their decisions benefit oneself, even if they are unfair (Bocian & Wojciszke, 2014a).
Mind Perception	The extent to which an agent is attributed mental capacities, specifically agency (ability to think and act) and experience (capacity for emotions) (Gray et al., 2012).
Liking	Affective evaluation of the decision-maker, serving as a mediator between self-interest and moral judgments (Bocian et al., 2018).

We tested the following hypotheses:

1. The decider making a beneficial decision for the perceiver's interests but violating fairness norms will be judged as more moral (and their decisions as fairer) compared to the same unfair decision benefiting the other participant (H1).
2. AI would be liked less (H2a) and seen as having less of a mind (H2b) than humans.
3. In effect, self-interest would be weaker for AI than for human agents (H3).

In all studies, we also measured how strongly participants perceived the decisions of AI and human agents as fair. Past research found evidence that people perceived high and medium financial outcomes as fair, even if they resulted from unfair procedures (Greenberg, 1987). Correspondingly, people judge overpayment as fair when it regards themselves and unfair when it concerns similar others (Greenberg, 1983). We hypothesized that people would judge the self-beneficial unfair decisions as fair and other-beneficial decisions as unfair (Studies 1 & 2). Moreover, this allows for differentiation between judgments of fairness of the decision and judgments of the moral character of the agent. As previous studies mostly focused on fairness and justice of the decision itself (for review, see Starke et al., 2022), we extended the literature by examining the agent's moral character. Thus, fairness of the decision would be a second indicator of biased moral cognition, which might help explain the differences between AI and human decision-makers.

1.4. Study 1

In Study 1, we sought to examine whether self-interest biases moral character and fairness judgments of human and AI decision-makers. We hypothesized that the decider making a beneficial decision for the perceiver's interests but violating fairness norms will be judged as more moral (and their decisions as fairer) compared to the same unfair decision benefiting the other participant (H1). Moreover, we hypothesized that AI should be liked less than human agents (H2a), and thus, the effect of self-interest would be weaker for AI than for human agents (H3).

2. Method

In this article, we report all manipulations and any data exclusions. Any additional measures not included in the primary analyses are reported in the Supplementary Materials. The relevant Research Ethics Committees have approved all studies. All participants provided informed consent. All raw data files, analysis scripts, and materials used in this article are available for download from the Open Science Framework (OSF; Data are available at https://osf.io/qpkaj/).

Participants. As estimating the effect size is difficult in the novel context, we took the 50th percentile effect size in social psychology established by Lovakov and Agadullina (2021), which is $d = 0.36$. We calculated the target sample size using G*Power (Faul et al., 2009) as $N = 341$ (assuming the power of 0.80). Considering possible exclusions, we aimed to recruit 400 participants using Prolific Academic. In the end, we managed to recruit 404 participants from the US. We excluded 72 participants who either failed an attention check or did not complete the study. While this reduced our final sample to $N = 332$ (167 identifying as women, 160 as male, and 5 identifying otherwise or preferring not to provide this information, $M_{age} = 41.36$, $SD_{age} = 12.39$), a post-hoc power analysis confirmed that our study retained 80 % power to detect an effect size of f = 0.18, which aligns with standard effect sizes in moral judgment research (Lovakov & Agadullina, 2021).

Design and procedure. Participants were randomly assigned to one of four experimental conditions in a 2 (decision: self-beneficial vs. other-beneficial) × 2 (decision-maker: human vs. AI) between-subjects design. First, we told the participants that we would pair them with a decider and another participant. The decider was either an AI algorithm or another participant, who would later decide how to distribute a £1 bonus between them and another participant. The bonus was additional money; each participant received a base payment of £1 for their time spent on the study.

To enhance realism and ensure participants perceived the AI or human decision-maker as legitimate, we employed a simulated interaction. Prior research highlights that participants may disengage from studies where AI involvement appears artificial or transparent (Wilson et al., 2022). Thus, the chat room was pre-coded with automated responses to create the impression of genuine human interaction. Given that deception is commonly used in experimental social psychology to simulate realistic decision-making contexts, we ensured compliance with APA ethical guidelines by debriefing participants entirely at the study's conclusion. From the debriefing, we can confirm that manipulation and debriefing were successful. Moreover, deception was necessary to control for confounds in decision-making expectations, ensuring that participants viewed both AI and human decision-makers as equivalent in decision authority. All manipulation checks for all studies can be found in the Supplementary Materials.

Further, we asked participants to complete a short writing task to write as many words correctly as possible. The task displayed sentences that changed every 10 s. Participants had to rewrite these sentences, with each word disappearing as they pressed the spacebar. Participants who had an accuracy below 5 % were removed after the main task before answering the dependent variable questions. The task lasted for 1 min. After the task, the participants saw a page with their accuracy, their paired participant's accuracy, and the decision outcome (see Fig. 1). Their accuracy was real, and the mock participants' accuracy was coded as 12 % more or less than theirs, depending on the research condition.

Participants saw that the decision-maker (AI or human) split the money unfairly, but depending on the research condition either beneficially for them or beneficially for the other participant. The decision was always unfair, meaning the person with better accuracy in the task got less money. The decision outcome, however, was either beneficial to the participant (they did worse but got more money, £0.75) or not (they did better but got less money, £0.25). After the decision, we asked participants about their impressions of the decision-maker and the decision. The median time spent on the study was 6 min 42 s.

3. Measures

Liking of the decider was measured with two items: "I like this human decider/AI decider" and "I would like to interact with this human decider/AI decider in the future". Participants indicated to what extent they agreed with each statement using a scale from 1 = *strongly disagree* to 7 = *strongly agree*. The items were averaged to create a composite score of liking the decider ($\alpha = .93$, $M = 3.70$, $SD = 2.07$).

Moral character judgments of the decider were measured with five items adjusted from Abele et al. (2016): "This human/AI decider is just", "This human/AI decider is moral", "This human/AI decider is trustworthy", "This human/AI decider is reliable", "This human/AI decider is

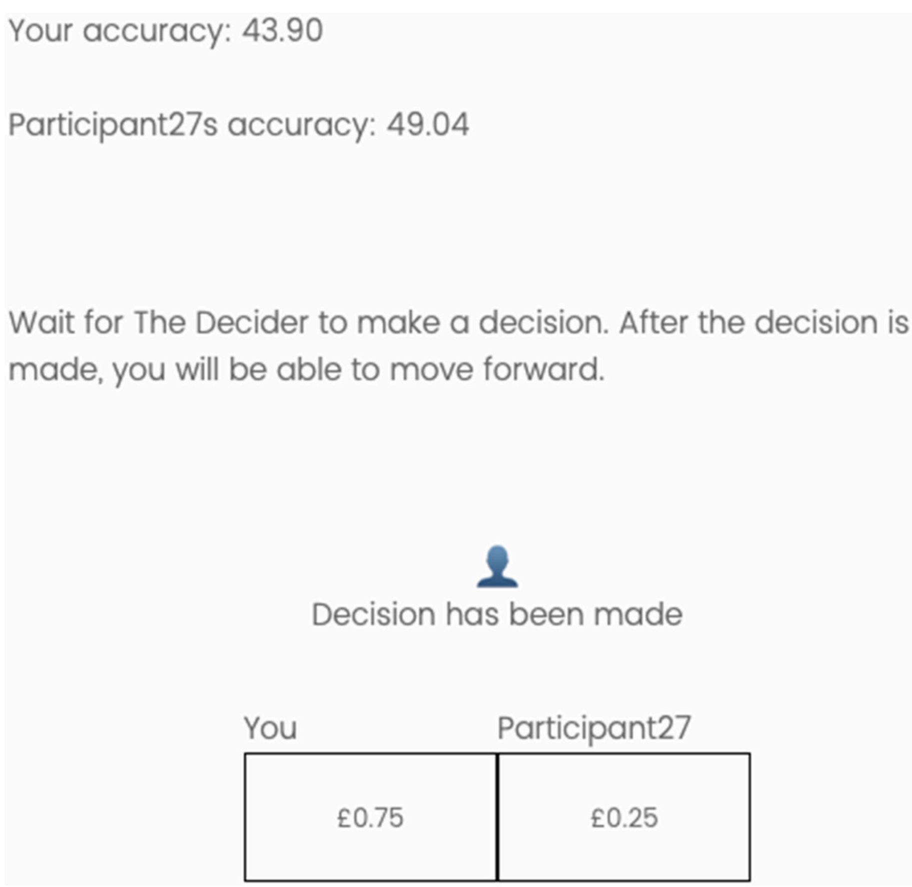

Fig. 1. *Presentation of the decision outcome to the participants (Study 1)*
Note. The figure is a screenshot of the study on the self-beneficial human condition. Participants' accuracy is their real accuracy, and another participant's accuracy is higher than theirs, but the other participant got a smaller bonus. Hence, the decision is unfair but beneficial to the participant.

honest". Participants indicated to what extent they agreed with each statement using a scale from 1 = *strongly disagree* to 7 = *strongly agree*. The items were averaged to create a composite score of perceived moral character (α = .96, M = 3.16, SD = 1.55).

Fairness of the decision was measured with one item: "How fair was the decision?". As indicated by the review by Starke et al. (2022) one-item fairness assessment is a common practice. Participants answered on a scale from 1 = *not at all* to 7 = *extremely* (M = 2.46, SD = 1.53).

4. Results

Table 2 presents the correlations of the main variables. The moral character was positively associated with liking and fairness perception. We have performed a 2 (decision: self-beneficial vs. other-beneficial) x 2 (decision maker: human vs. AI) between-participants ANOVA on each dependent variable. Means and standard deviations for all conditions can be found in Table 3.

Liking. We found a main effect of decision outcome. The decider was liked more when the decision was self-beneficial (M = 5.33, SD = 1.38), compared to other-beneficial (M = 2.17, SD = 1.29). However, the main effect of the decider and the interaction were insignificant (see Table 4).

Moral character judgments. We found a main effect of the decision outcome (see Table 4). Moral character judgments were higher when the decision was self-beneficial (M = 4.07, SD = 1.37), compared to other-beneficial (M = 2.31, SD = 1.18). However, we did not find differences in the perception of moral character between the AI and human decider, as the main effect of the decider was insignificant, neither the interaction between the decider and the decision (see Table 4).

Fairness. We found a main effect of the decision outcome. When the decision was self-beneficial, it was perceived as fairer than when the decision was other-beneficial (M = 3.05, SD = 1.57 vs. M = 1.90, SD = 1.26). We did not find the main effect of the decider, nor the interaction (see Table 4).

5. Discussion

Study 1 confirmed our first hypothesis (H1) that the decider whose unethical decision benefited participants would be judged as more moral (and their decision as fairer) than the decider whose unethical decision benefited another participant. We did not confirm H2a and H3, as there were no differences in participants' liking and moral character judgments between human and AI decision-makers. We examined secondary variables, including perceived intentionality, to assess whether the null effect could be attributed to failed manipulation. Participants rated decisions made by human agents as more intentional than those produced by AI, consistent with previous research (Wilson et al., 2022). This suggests that participants differentiated between human and AI decision-makers, supporting the credibility of the manipulation rather than explaining the null effect itself (see Supplementary Materials).

We recognize that the study is not without its limitations. Firstly, due to the unexpected difficulty of the main task, we had to remove more people than anticipated. According to the power analysis, the Study 1 sample provided a 0.80 power to detect the effect of f = 0.18, while the impact of the decider (AI vs. Human) on moral character judgments was f = 0.10. This suggests that the sample was underpowered to detect the main effect of the decider. Secondly, the accuracy rating was the participants' actual accuracy, on which the mock participants' accuracy was dependent, meaning that each participant received different feedback about their accuracy. While this is more ecologically accurate, it may have impacted participants' perceptions, as the mock participant's scores differed across participants. This further limits the self-interest manipulation, as if in the self-beneficial condition, adding 11.7 % to the participant's accuracy would have resulted in a value exceeding 100 %, the partner's score had to be capped at 100 % logically. While this still holds the unfair distribution assumption, thus holding the self-interest manipulation, it introduces unnecessary variation (this applied to six participants; excluding them does not change the pattern of results). Lastly, the bonus was additional money, and participants did get a base payment for the study regardless of the decision outcome. As an additional amount of money is positive, this could limit the strength of the effect. We address these limitations in Study 2.

Table 2
Table of correlations for main variables (study 1).

Variables	1	2	3
1. Liking	–		
2. Moral Character	0.78***	–	
3. Fairness	0.59***	0.76***	–

Note. ***p < .001.

Table 3
Means and standard deviations for the main variables (Study 1).

		AI	Human
		M (SD)	M (SD)
Liking	Self-beneficial	5.22 (1.52)	5.43 (1.24)
	Other- beneficial	2.02 (1.24)	2.31 (1.34)
Moral character	Self-beneficial	3.96 (1.42)	4.19 (1.32)
	Other- beneficial	2.17 (1.11)	2.44 (1.23)
Fairness	Self-beneficial	2.94 (1.45)	3.16 (1.68)
	Other- beneficial	1.84 (1.24)	1.97 (1.29)

5.1. Study 2

In Study 2, we aimed to address the limitations of Study 1 by conducting a methodologically improved and well-powered experiment to test our hypotheses. First, because people are strongly averse to losing (Kahneman & Tversky, 1979), we strengthen the self-interest manipulation by telling participants that they might lose some of their £1 bonus. Moreover, by penalizing participants for their mistakes, we wanted to activate unfair economic harm, which is universally recognized as immoral (Haidt, 2007). Therefore, we hypothesized that participants would perceive decision-makers who aided their interests as moral (and their decision as fairer) and those who benefited others as immoral (and their decision as unfair) (H1).

Second, according to the Mind Perception Theory (Gray et al., 2012), perceiving an agent as having a "mind" is essential to making moral judgments. Therefore, the extent to which participants perceive the decision-maker as possessing a mind should influence their evaluations of that agent's moral character. Hence, we added mind perception as an additional measure to confirm that the AI and human decision-makers are perceived differently regarding mind qualities (e.g., Bigman & Gray, 2018) and further tested the relationship between the perceived mind of the decision-maker and moral character judgments. We presumed that the AI decision-maker would be liked less (H2a) and perceived as having less mind than the human decision-maker (H2b), which would explain why the self-interest impact on moral character judgments is weaker for the AI decision-maker and stronger for the human decision-maker (H3). This study was preregistered at https://aspredicted.org/WW6_6BS.

6. Method

Participants. In Study 1, the effect of self-interest on moral character judgments was f = 0.70. Meanwhile, the effect of the decider on moral character judgments was f = 0.09. We calculated that with a power of 0.80 for the interaction effect, we would need a total sample of N = 404. As we expected an interaction between self-interest and decider to show a reversal, according to Giner-Sorolla's (2018) powering guide, we then doubled the sample size, resulting in a total of N =

Table 4
Fixed-Effects ANOVA results for the Main Variables (Study 1).

Variable	Predictor	*df*	Mean Square	*F*	*p*	η^2p	η^2p 95 % CI [LL, UL]
Liking	Decider	1	5.39	3.01	.084	0.01	[0.00, 0.03]
	Decision	1	827.83	463.32	<.001	0.59	[0.53, 0.63]
	Interaction	1	0.14	0.08	.778	0.00	[0.00, 0.01]
Moral Character	Decider	1	5.09	3.15	.077	0.01	[0.00, 0.03]
	Decision	1	259.23	160.36	<.001	0.33	[0.26, 0.39]
	Interaction	1	0.03	0.02	.900	0.00	[0.00, 0.00]
Fairness	Decider	1	2.58	1.28	.259	0.00	[0.00, 0.02]
	Decision	1	109.43	54.21	<.001	0.14	[0.09, 0.20]
	Interaction	1	0.18	0.09	.765	0.00	[0.00, 0.01]

Note. LL and UL represent the lower-limit and upper-limit of the η^2p confidence interval, respectively.

808 participants. Thus, we aimed to recruit 810 participants using Prolific. In the end, we recruited 819 US participants. We then filtered out 62 for not finishing the study and/or not following the attention check. In contrast to Study 1, participants who wrote less than 5 words in total in the writing task were removed before answering the dependent variable questions, thus not finishing the study fully. The final sample consisted of $N = 757$ participants (374 identifying as women, 371 as male, and 12 identifying otherwise or preferring not to provide this information; mean age = 43.79, $SD = 14.13$). Post hoc sensitivity analysis revealed that the sample size allows the detection of the effect size $f = 0.12$ with 0.80 power.

Design and procedure. Participants were randomly assigned to one of the four between-subject conditions based on scheme 2 (decision: self-beneficial vs. other-beneficial) x 2 (decision maker: human vs. AI). Study 2 was similar to Study 1 with the following changes. First, participants were told that they start with a £1 bonus. The bonus was additional money; each participant received a base payment of £0.80 for their time spent on the study. The writing task was the same as in Study 1, with an additional 10s practice round. We also made the cut-off for participants more lenient, and they were removed from the study if they had written less than 5 words in total, regardless of the amount of mistakes they made.

In contrast to Study 1, the feedback was hard-coded, so the participant always had a 23.4 % mistake rate, and depending on the research condition, the other participant had either a 12.1 % or 34.6 % mistake rate. We would like to note a minor inaccuracy here, as the feedback difference differs by 0.1 %. In projects replicating this method we suggest other authors to adjust the difference to match exactly. Participants real mean mistake rate in this study was 20.30 %, which is very close to the fake value provided in the study. Finally, compared to Study 1, participants did not gain their bonus but could lose it. As in Study 1, the outcome of the decision was always unfair but either self-beneficial to the participant or beneficial to the other participant. In other words, participants made either more mistakes but got a smaller penalty (-£0.25

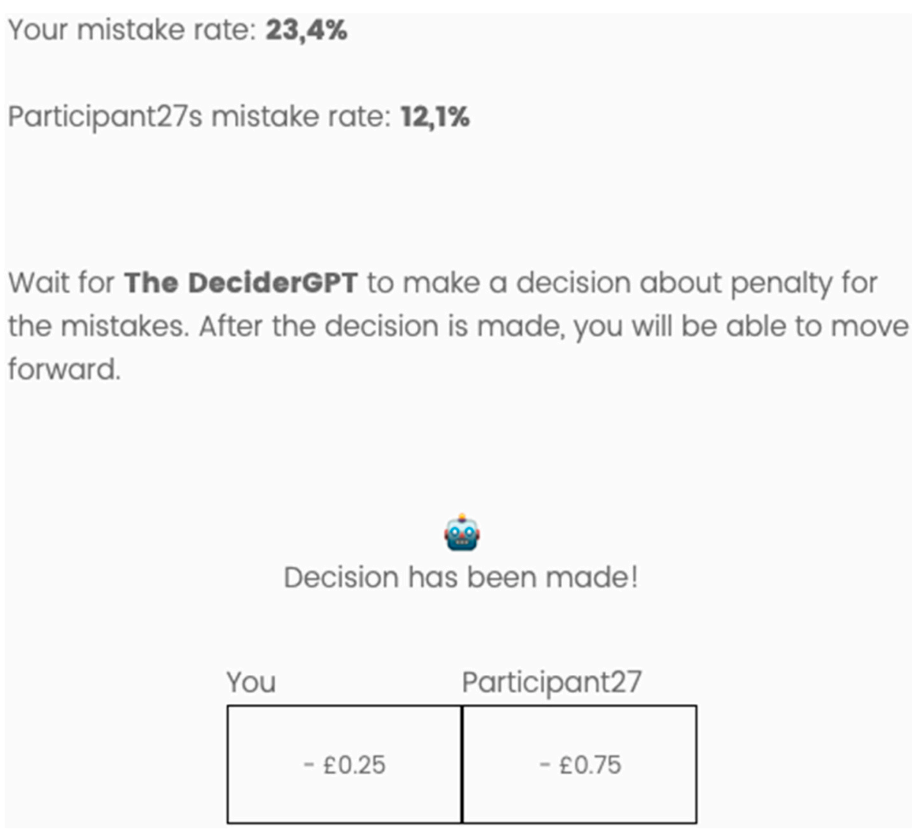

Fig. 2. *Presentation of the decision outcome to the participants (Study 2).*
Note. The image is a screenshot from the study of the self-beneficial AI condition. On the same slide, participants could see both their and other participants mistake rate and the decision about the penalty for them and another participant. The participant was told they made more mistakes but got a smaller penalty, hence the self-beneficial outcome.

vs -£0.75) or made fewer mistakes but got a bigger penalty (-£0.75 vs -£0.25; see Fig. 2). The median time spent on this study was 8 min 31 s.

7. Measures

Liking of the decider was measured as in Study 1 (α = .93, $M = 3.21$, $SD = 1.94$).

Mind perception of the decider was measured with a six-item scale from Bigman and Gray (2018). Three items measured agency: "This AI/human decider can communicate with others", "This AI/human decider is capable of thinking", "This AI/human decider plans its actions," and three experience: "This AI/human decider is sensitive to pain", "This AI/human decider can experience happiness", "This AI/human decider can experience fear". Participants indicated to what extent they agreed with each statement using a scale from 1 = *strongly disagree* to 7 = *strongly agree*. Averaged results for both subscales indicate perceived mind (α = .92, $M = 3.76$, $SD = 1.87$).

Moral character judgments of the decider were measured as in Study 1 (α = .96, $M = 3.21$, $SD = 1.73$).

Fairness of the decision was measured as in Study 1 ($M = 2.89$, $SD = 1.96$).

8. Results

Table 5 presents the correlations of the main variables. The moral character was positively associated with liking, fairness and mind perception. We performed a 2 (decision: self-beneficial vs. other-beneficial) x 2 (decision maker: human vs. AI) between-subjects ANOVA on each dependent variable to test our predictions. Means and standard deviations for all conditions can be found in Table 6.

Liking. As in Study 1, we found a main effect of the decision, with the target who makes a decision beneficial to the participant being liked more than the target who makes an other-beneficial decision ($M = 4.32$, $SD = 1.64$ vs. $M = 2.14$, $SD = 1.58$). This time, we also found a main effect of the decider, with the human being more liked than AI ($M = 3.39$, $SD = 1.90$ vs. $M = 3.03$, $SD = 1.97$). The interaction was nonsignificant (see Table 7).

Mind perception. As expected, we found a main effect of the decider. Participants rated human decision-makers higher than AI on overall perceived mind ($M = 5.11$, $SD = 1.45$ vs. $M = 2.43$, $SD = 1.16$). Interestingly, we have also found a main effect of the decision. The decision-maker who made the self-beneficial decision was attributed greater perceived mind, compared to the other-beneficial decision ($M = 4.07$, $SD = 1.83$ vs. $M = 3.46$, $SD = 1.87$). The interaction was insignificant (see Table 7).

Moral character. We found the main effect of the decision. As hypothesized, the decision-maker was perceived as more moral when the decision was self-beneficial, compared to other-beneficial ($M = 4.00$, $SD = 1.52$ vs. $M = 2.43$, $SD = 1.57$). Further, we found a main effect of the decider, with the human decision-maker being attributed more moral character than the AI ($M = 3.56$, $SD = 1.78$ vs. $M = 2.86$, $SD = 1.61$). Additionally, we found an interaction between the decision and the decider (See Fig. 3).

The pairwise comparison revealed that, as hypothesized, while in both human and AI conditions, the other-beneficial agent was judged as less moral than the self-beneficial agent, the effect of self-interest on moral character judgments was stronger for the human decision-maker ($t(367.59) = -12.09$, $p < .001$, $d = -1.25$, 95 % CI [−1.47, −1.03]) than AI decision maker ($t(379) = -8.11$, $p < .001$, $d = -0.83$, 95 % CI [−1.04, −0.62]). Finally, AI was judged as less moral than humans in both the self-beneficial condition ($t(371) = -6.84$, $p < .001$, $d = -0.71$, 95 % CI [−0.92, −0.50]) and the other-beneficial condition ($t(382) = -2.29$, $p = .023$, $d = -0.23$, 95 % CI [−0.43, −0.03]).

Table 5
Table of correlations for main variables (study 2).

Variables	1	2	3
1. Liking	–		
2. Mind	0.38***	–	
3. Moral Character	0.81***	0.52***	–
4. Fairness	0.70***	0.39***	0.80***

Note. ***p < .001.

Table 6
Means and standard deviations for the main variables (Study 2).

		AI	Human
		M (*SD*)	*M* (*SD*)
Liking	Self-beneficial	4.14 (1.75)	4.48 (1.52)
	Other- beneficial	1.98 (1.54)	2.31 (1.60)
Mind	Self-beneficial	2.65 (1.12)	5.47 (1.21)
	Other- beneficial	2.23 (1.17)	4.74 (1.57)
Moral character	Self-beneficial	3.49 (1.46)	4.51 (1.41)
	Other- beneficial	2.26 (1.52)	2.62 (1.61)
Fairness	Self-beneficial	3.29 (1.79)	4.11 (1.91)
	Other- beneficial	2.05 (1.66)	2.14 (1.68)

Fairness. We found an interaction effect between the decision and the decider (see Fig. 4). Pairwise comparisons showed that, in the self-beneficial condition, AI was judged as less fair than a human making the same decision ($t(371) = -4.27$, $p < .001$, $d = -0.44$, 95 % CI [−0.65, −0.24]). In the other-beneficial condition, the difference was insignificant ($t(382) = -0.54$, $p = .588$). For both AI and human deciders, the self-beneficial decision was seen as fairer than the other-beneficial decision, however, the effect was stronger for the human agent ($t(368.03) = -10.63$, $p < .001$, $d = -1.10$, 95 % CI [−1.31, −0.88]) compared to the AI agent ($t(379) = -7.05$, $p < .001$, $d = -0.72$, 95 % CI [−0.93, −0.51]).

Moderated mediation analyses for liking and mind perception. To test our hypothesis that the effect of self-interest on moral character judgments is driven by liking and that liking explains the difference between human and AI decision-makers, we ran a moderated mediation with independent variable as decision outcome (X), dependent variable as moral character (Y), liking as a mediator (M) and decider as a moderator (W; see Fig. 5). Variables were effect coded, for the decision outcome (Self-beneficial = 1 and other-beneficial = −1) and the decider (Human = 1 and AI = −1)

We based the mediation model on the previous studies, which showed that liking mediates the effect of self-interest on moral character judgments (Bocian & Wojciszke, 2014a) as well as on the theoretical assumptions that affective changes toward a transgressor drive self-interest bias (Bocian et al., 2020). We used Model 15 from the PROCESS macro (Hayes, 2013), which tests a moderated mediation model where a fourth variable moderates the mediator-to-outcome path. This model was chosen based on the significance of moderation analysis on each path in the model (see Supplementary Materials for the complete analysis).

We confirmed that the effect of self-interest on moral character judgments through liking was moderated by who makes the decision ($B = 0.11$, $SE = 0.06$, 95 % CI = [0.01, 0.22]). The indirect effect was significant for both the AI decider ($B = 0.73$, $SE = 0.05$, 95 % CI = [0.63, 0.83]), and the human decider ($B = 0.84$, $SE = 0.06$, 95 % CI = [0.73, 0.96]), however, it was stronger for the human decider ($B = 0.11$, $SE = 0.06$, 95 % CI = [0.01, 0.22]). The alternative moderation model with the moral character as a mediator and liking as the dependent variable was insignificant ($B = 0.00$, $SE = 0.04$, 95 % CI = [−0.08, 0.09]).

These results suggest that liking serves as a cognitive-affective bridge between self-interest and moral judgment. When a decision-maker acts in a way that benefits the observer, affective responses (liking) bias moral evaluations, making the agent appear more ethical. This finding is consistent with research on the mere liking effect in moral cognition, which shows that positive interpersonal attitudes shape moral character attributions independent of objective behavior (Bocian et al., 2018).

Table 7
Fixed-Effects ANOVA results for the Main Variables (Study 2).

Variable	Predictor	*df*	Mean Square	*F*	*p*	η^2p	η^2p 95 % CI [LL, UL]
Liking	Decision	1	892.56	347.48	<.001	0.32	[0.27, 0.36]
	Decider	1	21.20	8.26	.004	0.01	[0.00, 0.03]
	Interaction	1	0.01	0.00	.959	0.00	[0.00, 0.00]
Mind	Decision	1	62.02	38.00	<.001	0.05	[0.03, 0.08]
	Decider	1	1348.08	826.07	<.001	0.52	[0.49, 0.56]
	Interaction	1	4.77	2.92	.088	0.00	[0.00, 0.02]
Moral Character	Decision	1	461.54	204.96	<.001	0.21	[0.17, 0.26]
	Decider	1	90.15	40.03	<.001	0.05	[0.03, 0.08]
	Interaction	1	20.01	8.89	.003	0.01	[0.00, 0.03]
Fairness	Decision	1	490.14	157.87	<.001	0.17	[0.14, 0.21]
	Decider	1	39.35	12.67	<.001	0.02	[0.01, 0.04]
	Interaction	1	25.02	8.06	.005	0.01	[0.00, 0.03]

Note. LL and UL represent the lower-limit and upper-limit of the η^2p confidence interval, respectively.

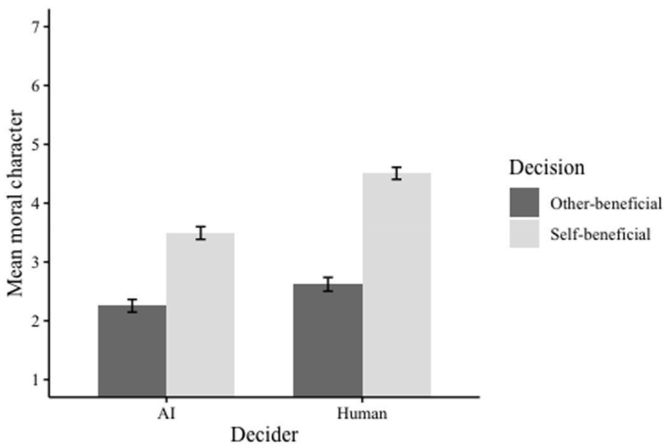

Fig. 3. *Mean moral character judgments as a function of decision and decider (Study 2).*
Note. The error bars represent one standard error.

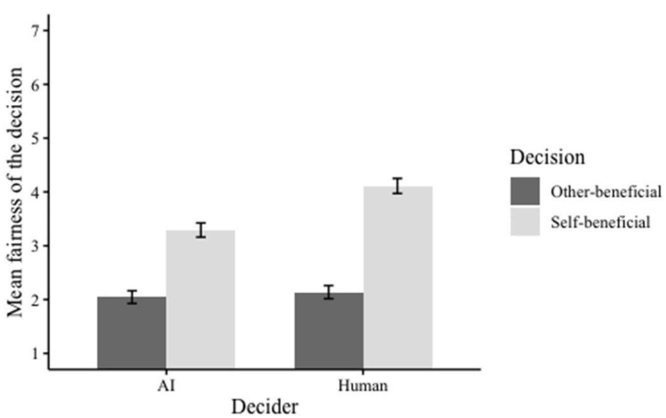

Fig. 4. *Mean fairness judgments as a function of decision and decider (Study 2).*
Note. The error bars represent one standard error.

Significantly, this effect was attenuated for AI decision-makers, suggesting that AI is perceived as less subject to emotional engagement, which weakens self-interest biases in moral judgments.

Further, according to Mind Perception Theory (Gray et al., 2012), moral judgments should be explained by the perceived mind of the decider. This was further developed by Bigman and Gray (2018), who showed that mind perception mediates the effect of the agent on permissibility. To explore this reasoning, we ran a moderated mediation with the independent variable as the decider (X), the dependent variable as moral character (Y), mind perception as a mediator (M), and self-beneficial vs. other-beneficial decision as a moderator (W; see Fig. 6). Variables were effect coded for the decision outcome (Self-beneficial = 1 and Other-beneficial = −1) and for the decider (Human = 1 and AI = −1). We run Model 5 in the PROCESS macro (Hayes, 2013) based on the significance of moderation analysis on each path in the model (see Supplementary Materials for the complete analysis). As expected, the decider significantly predicted perceived mind (B = 1.34, SE = 0.05, $p < .001$, 95 % CI = [1.24, 1.43]) indicating that human decision-makers were attributed greater percieved mind than AI decision-makers. In turn, mind perception positively predicted moral character (B = 0.61, SE = 0.04, $p < .001$, 95 % CI = [0.53, 0.68]). The indirect effect of the decider on moral character judgments through perceived mind was significant ($B = 0.81$, $SE = 0.05$, 95 % CI = [0.70, 0.92]). However, the alternative model, with mind perception as the dependent variable and the moral character as a mediator, was insignificant ($B = -0.01$, $SE = 0.04$, 95 % CI = [−0.07, 0.08]). Further, the direct effect of the decider type on moral character ratings was moderated by the nature of the decision ($B = 0.11$, $SE = 0.05$, $p = .015$, 95 % CI [0.02, 0.21]). When the decision was other-beneficial, there was a significant negative direct effect, indicating lower moral character ratings for human deciders compared to AI, controlling for mind perception ($B = -0.58$, $SE = 0.08$, $p < .001$, 95 % CI [−0.74, −0.42]). This negative direct effect persisted when the decision was self-beneficial, but it was significantly weaker ($B = -0.35$, $SE = 0.08$, $p < .001$, 95 % CI [−0.52, −0.18]). This pattern suggests a suppression effect, where the positive indirect pathway through perceived mind obscures a negative direct effect of being a human decider on moral character judgments. In other words, humans appear more moral overall primarily because they are perceived as having more mind. However, when controlling for this perception, the direct comparison reveals that AI deciders are actually rated as more moral than human deciders.

In sum, we confirmed the hypothesis (H3) that the impact of self-interest on moral character judgments was weaker for the AI decision-maker because AI was liked less (H2a) and perceived as having less mind (H2b) than the human decision-maker. As predicted by the Mind Perception Theory (Gray et al., 2012), our results show that moral judgments of decision-makers are shaped by perceptions of their mental capacities—particularly their agency (ability to act intentionally) and experience (capacity for emotions). AI decision-makers were perceived as lower in both dimensions, reducing their moral character ratings compared to human agents. This aligns with prior findings that people are less likely to hold AI morally accountable due to perceived deficits in agency and experience (Bigman & Gray, 2018).

9. Discussion

Study 2 confirmed all our hypotheses. As we predicted, the decision-maker who made the unfair decision that benefited participants was judged as more moral (and their decision as fairer) and liked more than

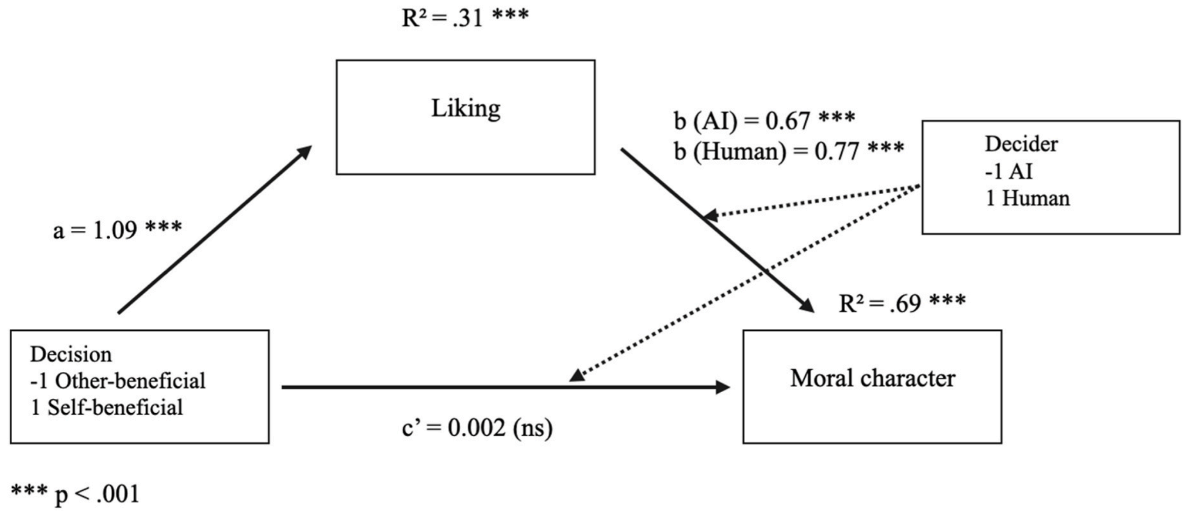

Fig. 5. Model for the moderated mediation of the effect of self-interest on moral character judgments through liking (Study 2).

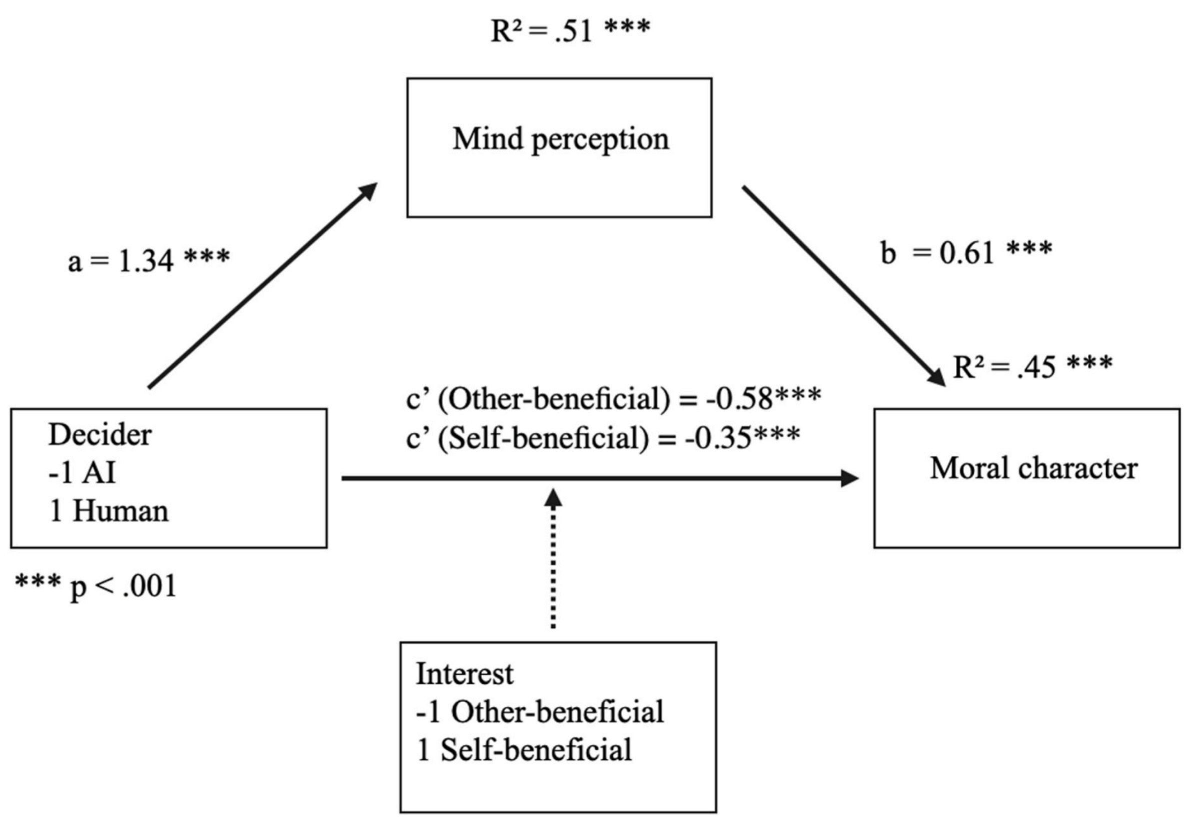

Fig. 6. Model for the moderated mediation of the effect of the agent on moral character judgments through mind perception (Study 2).

the one who made the unfair and financially harmful decision for participants. Further, this effect was smaller for the AI decision-maker than for the human decision-maker because AI was liked less and perceived as having less of a mind than humans. To assess whether the effect was robust to individual differences in performance perceptions, we conducted an exploratory analysis including typing accuracy, mistakes made and words typed as covariates. The key Decider × Self-interest interactions remained significant for moral character judgements (F(1, 750) = 11.17, $p < .001$, η^2p = .02 [0.004, 0.32]), and fairness (F(1, 750) = 9.60, $p = .002$, η^2p = .01 [0.003, 0.29]; see Supplementary Materials for full model).

Overall, the results confirm that if people benefit from an unfair decision, self-interest biases their moral character (and fairness) judgments for both human and AI decision-makers. However, this effect is more robust when a human makes the decision than an AI because people like AI agents less than human agents and believe they have fewer mental qualities. Therefore, the presented evidence implies that people's moral character judgments are less biased by their interests when AI than when humans represent decision-makers. Still, it also confirms that the judgments of AI decision-makers are not free from self-interest bias. Importantly, our results suggest that self-interest not only biases moral judgments, but also affects perceptions of mind: participants attributed more mind to decision-makers whose choices benefited them, indicating a novel self-serving cognitive distortion in mind perception.

9.1. Study 3

In Study 3, we sought to generalize evidence for our hypotheses. Therefore, first, we aimed to replicate the effects of self-interest bias, liking, and mind perception found in Study 2. Second, to test the robustness of the self-interest bias, we decided to compare the moral character judgments of decision-makers who submit unfair but beneficial decisions with the fair but financially disadvantageous decisions for participants. Hence, we introduced a control condition scenario instead of the "other-beneficial" scenario from Study 2. In the control condition, the decision-maker made a fair decision: the participant had a higher mistake rate than the other player, and effectively they received the larger penalty, resulting in more money being taken from them (see Fig. 7). This study was preregistered at https://aspredicted.org/Q91_M5W.

10. Method

Participants. In Study 2, the interaction effect between the decider and self-interest on moral character judgments was $f = 0.11$. We calculated using the G*Power Calculator (Faul et al., 2009) that to replicate this effect with a power of 0.80, we would need a total sample of $N = 649$. Considering possible exclusions, we aimed to recruit 700 participants from Prolific. In the end, we recruited $N = 862$. We then filtered out 71 participants for not finishing the study and/or not passing the attention check. The final sample consisted of $N = 791$ US participants, (385 identifying as women, 390 as men and 16 identifying otherwise or preferring not to provide this information; $M_{age} = 42.84$, $SD_{age} = 13.37$). Post hoc sensitivity analysis revealed that the sample size allows the detection of the effect size $f = 0.10$ with 0.80 power.

Design and procedure. Participants followed the same procedure as in Study 2, with two exceptions. First, we used a control condition instead of the "other-beneficial" scenario. In the control condition, the other responder had a 12.1 % mistake rate, and the main participant made more mistakes and received a bigger penalty (-£0.75 vs. -£0.25). In other words, in the control condition, the decision was fair. Still, participants lost more money, in contrast to the self-beneficial condition, where the decision was unfair, but participants lost less money. Similarly to Study 2, actual mean mistake rate of participants was 21.33 %, very close to the 23.4 % told them in the study. As in previous studies, the bonus was additional money; each participant received a base payment of £0.9 for their time spent on the study. The median time spent on the study was 08 min and 51 s.

11. Measures

Liking was measured as in Study 1 (α = .91, $M = 3.64$, $SD = 1.83$).

Mind perception of the target was measured as in Study 2 (α = .92, $M = 3.99$, $SD = 1.87$).

Moral character judgments were measured as in Study 1 (α = .93, $M = 3.84$, $SD = 1.59$).

Fairness of the decision was measured as in Study 1 ($M = 3.59$, $SD = 1.96$).

12. Results

Table 8 presents the correlations of the main variables. The moral character was positively associated with liking, fairness and mind perception. As preregistered, we performed a 2 (decision: self-beneficial vs. fair) x 2 (decision maker: human vs. AI) between-subjects ANOVA on

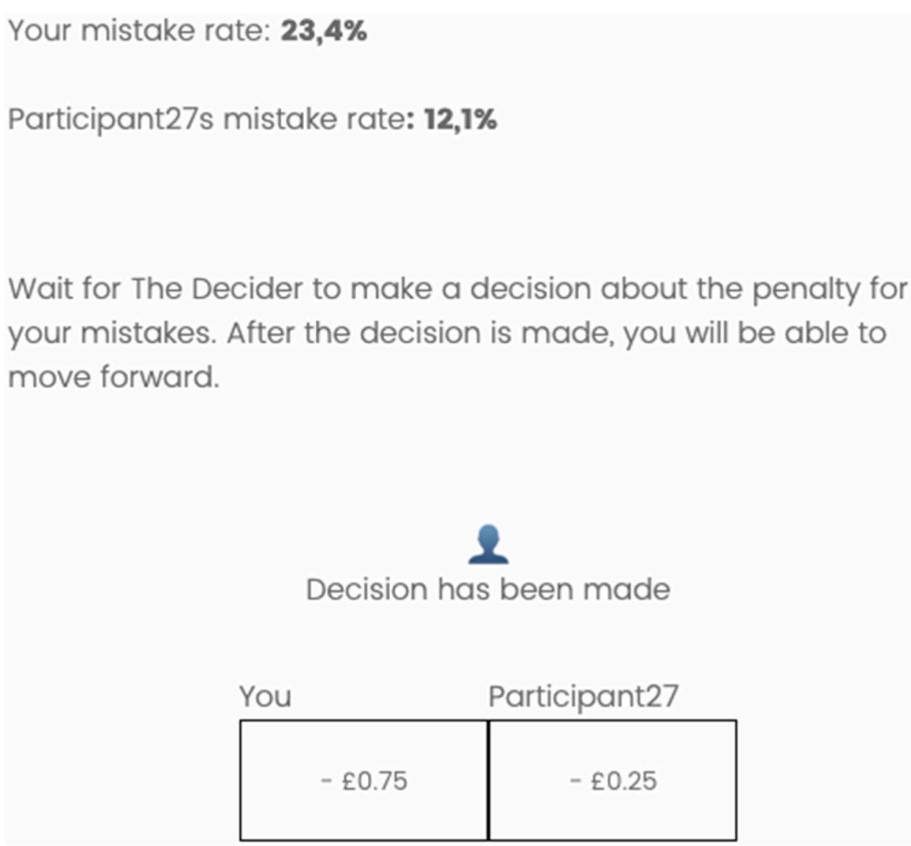

Fig. 7. *Presentation of the decision outcome to the participants (Study 3).*
Note. The image is a screenshot from the study of the fair human condition. The participants were told they made more mistakes and got a bigger penalty, so the decision was fair.

Table 8
Table of correlations for main variables (study 3).

Variables	1	2	3	4
1. Liking	–			
2. Mind	0.42***	–		
3. Moral Character	0.72***	0.57***	–	
4. Fairness	0.58***	0.29***	0.70***	–

Note. ***p < .001.

each dependent variable. Means and standard deviations for all conditions can be found in Table 9.

Liking. Replicating the results found in Study 2, we found the main effect of decider. A human decider was liked more than an AI decider ($M = 3.91$, $SD = 1.81$ vs. $M = 3.39$, $SD = 1.81$; see Table 10). Replicating the Study 1 and 2, we found the main effect of the decision. With the decider who made the self-beneficial decision being liked more than while making a fair decision ($M = 4.18$, $SD = 1.71$ vs. $M = 3.11$, $SD = 1.78$. The interaction was also significant (see Fig. 8).

Pairwise comparisons revealed that in the self-beneficial condition, the human was liked more than AI ($t(386.16) = -5.13$, $p < .001$, $d = -0.52$, 95 % CI [−0.72, −0.32]). In the fair condition, there was no difference between the liking of the human vs. AI ($t(396) = -0.89$, $p = .372$). Notably, comparing the effect of the decision on the liking across the decider manipulation, in both conditions the self-beneficial agent was liked more than the fair agent. Still, the effect was stronger for the human decider ($t(376.71) = -8.42$, $p < .001$, $d = -0.86$ 95 % CI [−1.06, −0.65]) compared to the AI decider ($t(402) = -4.11$, $p < .001$, $d = -0.41$, 95 % CI [−0.61, −0.21]).

Mind perception We found the main effect of the decider, with a human decider having more perceived mind than an AI decider ($M = 5.42$, $SD = 1.31$ vs. $M = 2.62$, $SD = 1.17$). However, we did not find the main effect of the decision or the interaction (see Table 10).

Moral character judgments. Results replicated the effects on moral character judgments found in Study 2. Similarly to Study 2, the decider making a self-beneficial decision was seen as having more moral traits than the decider making a fair decision ($M = 3.96$, $SD = 1.56$ vs. $M = 3.73$, $SD = 1.62$). Further, humans were judged as having more moral character than AI ($M = 4.30$, $SD = 1.52$ vs. $M = 3.40$, $SD = 1.53$). Further, we again found an interaction effect (see Fig. 9). The pairwise comparison revealed that AI was judged as less moral than humans in both the self-beneficial ($t(391) = -8.74$, $p < .001$, $d = -0.88$, 95 % CI [−1.09, −0.67]) and the fair condition ($t(396) = -3.38$, $p = .001$, $d = -0.34$, 95 % CI [−0.54, −0.14]). Further, in the human condition, although self-beneficial human was judged as more moral compared to the fair human, ($t(376.36) = -3.86$, $p < .001$, $d = -0.39$, 95 % CI [−0.59, −0.19]), the perceptions of AI decider did not differ across the conditions ($t(402) = 0.86$, $p = .390$).

Fairness. We found the main effect of the decider. The human-made decision was perceived as fairer overall than the decision made by the AI ($M = 3.77$, $SD = 1.93$ vs. $M = 3.42$, $SD = 1.98$). The main effect of the decision was insignificant, but the interaction effect between the decider and the decision was significant (see Fig. 10). Pairwise comparisons revealed that in the self-beneficial condition, the AI-made decision was seen as less fair than a human-made decision ($t(390.45) = -3.32$, $p = .001$, $d = -0.34$, 95 % CI [−0.53, −0.14]). In the fair condition, the difference was insignificant ($t(396) = -0.34$, $p = .734$). Further, in the human condition, the decision made by the self-beneficial human was seen as fairer than the fair human ($M = 4.04$, $SD = 1.83$ vs. $M = 3.50$, $SD = 2.00$; $t(381.80) = -2.76$, $p = .006$, $d = -0.28$, 95 % CI [−0.48, −0.08]). In the AI conditions, the fair and the self-beneficial AI did not differ in fairness ($t(402) = 0.14$, $p = .891$, $d = 0.01$, 95 % CI [−0.18, 0.21]).

Table 9
Means and standard deviations for the main variables (Study 3).

		AI	Human
		M (*SD*)	*M* (*SD*)
Liking	Self-beneficial	3.76 (1.77)	4.62 (1.54)
	Fair	3.03 (1.79)	3.19 (1.78)
Mind	Self-beneficial	2.60 (1.01)	5.52 (1.13)
	Fair	2.65 (1.32)	5.32 (1.46)
Moral character	Self-beneficial	3.34 (1.47)	4.60 (1.38)
	Fair	3.47 (1.59)	4.01 (1.60)
Fairness	Self-beneficial	3.41 (1.95)	4.04 (1.83)
	Fair	3.43 (2.01)	3.50 (2.00)

Moderated mediation analyses for liking and mind perception. As in Study 2, we ran a moderated mediation using model 8 from PROCESS (Hayes, 2013), with the independent variable as decision outcome (self-beneficial vs. fair) (X), the dependent variable as moral character (Y), liking as a mediator (M), and the decider as a moderator (W), see Fig. 11. Variables were effect-coded such that human = 1 and AI = −1 for the decider, and self-beneficial = 1 and fair = −1 for the decision outcome. The analysis confirmed that the entity of the decider moderated the effect ($B = 0.22$, $SE = 0.08$, 95 % CI = [0.07, 0.38]). The indirect effect of the decision outcome on moral character judgments through liking was again significant for both the human decider ($B = 0.45$, $SE = 0.06$, 95 % CI = [0.34, 0.57]) and the AI decider ($B = 0.23$, $SE = 0.06$, 95 % CI = [0.12, 0.34]). As in Study 2, the effect was stronger for the human decider ($B = 0.22$, $SE = 0.08$, 95 % CI = [0.07, 0.38]). The alternative moderation model with the moral character as a mediator and liking as the dependent variable was insignificant ($B = 0.03$, $SE = 0.04$, 95 % CI = [−0.05, 0.11]).

Further, as in Study 2, we ran a moderated mediation with the decider (AI vs Human) as the independent variable (X), the mind perception as a mediator (M), decision outcome (self-beneficial vs. fair) as the moderator (W), and moral character as the dependent variable (Y), see Fig. 12. Consistent with Study 2, the decider significantly predicted perceived mind ($B = 1.40$, $SE = 0.04$, $p < .001$, 95 % CI = [1.31, 1.48]). Specifically, participants attributed greater mind to human decision-makers than to AI. In turn, perceived mind significantly predicted moral character judgments ($B = 0.69$, $SE = 0.04$, $p < .001$, 95 % CI = [0.62, 0.76]). The indirect effect of the decider on moral character judgments through perceived mind was significant ($B = 0.97$, $SE = 0.05$, 95 % CI = [0.86, 1.07]). However, the alternative moderation model with the moral character as a mediator and mind perception as the dependent variable was insignificant ($B = -0.02$, $SE = 0.04$, 95 % CI = [−0.09, 0.05]). As in Study 2, the direct effect of the decider type on moral character ratings was moderated by the decision ($B = 0.14$, $SE = 0.04$, $p = .002$, 95 % CI [0.05, 0.22]). When the decision was fair, there was a significant negative direct effect, indicating lower moral character ratings for human deciders compared to AI, controlling for mind perception ($B = -0.65$, $SE = 0.08$, $p < .001$, 95 % CI [−0.81, −0.50]). This negative direct effect persisted when the decision was self-beneficial, but it was significantly weaker ($B = -0.38$, $SE = 0.08$, $p < .001$, 95 % CI [−0.54, −0.22]). Similarly to Study 2, these results suggest that humans appear more moral overall primarily because they are perceived as having more mind.

Taken together, we corroborated Study 2's results that the decision-maker's liking explains the influence of self-interest on moral character judgments and is stronger for humans than AI. Further, we replicated and confirmed that the different mind perceptions of AI and humans explain the impact of self-interest on moral character judgments.

13. Discussion

Study 3 replicated the results from Study 2, contrasting fair but disadvantageous decisions and unfair but beneficial decisions for participants. Even in the light of the fair decision, we found the robust effect of self-interest on moral character judgments, which as in Study 2, was stronger for the human than AI decision-maker. As in Study 2, the potential explanation of the difference between the human and AI

Table 10
Fixed-Effects ANOVA results for the Main Variables (Study 3).

Variable	Predictor	*df*	Mean Square	*F*	*p*	η^2p	η^2p 95 % CI [LL. UL]
Liking	Decision	1	228.70	77.00	<.001	0.09	[0.06, 0.12]
	Decider	1	51.144	17.22	<.001	0.02	[0.01, 0.04]
	Interaction	1	24.01	8.08	.005	0.01	[0.00, 0.03]
Mind	Decision	1	1.08	0.70	.402	0.00	[0.00, 0.01]
	Decider	1	1545.34	1004.74	<.001	0.56	[0.53, 0.59]
	Interaction	1	3.16	2.05	.152	0.00	[0.00, 0.01]
Moral Character	Decision	1	10.14	4.43	.036	0.01	[0.00, 0.02]
	Decider	1	159.78	69.78	<.001	0.08	[0.05, 0.11]
	Interaction	1	25.33	11.06	<.001	0.01	[0.00, 0.03]
Fairness	Decision	1	12.93	3.40	.066	0.00	[0.00, 0.02]
	Decider	1	24.39	6.41	.012	0.01	[0.00, 0.02]
	Interaction	1	15.81	4.16	.042	0.01	[0.00, 0.02]

Note. LL and UL represent the lower-limit and upper-limit of the η^2p confidence interval, respectively.

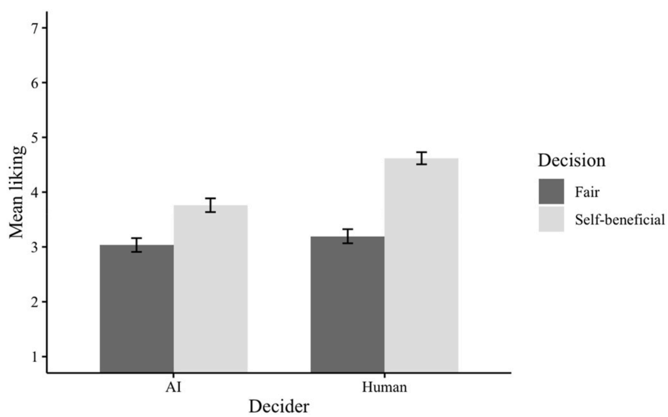

Fig. 8. Mean liking as a function of decision and decider (Study 3)
Note. The error bars represent one standard error.

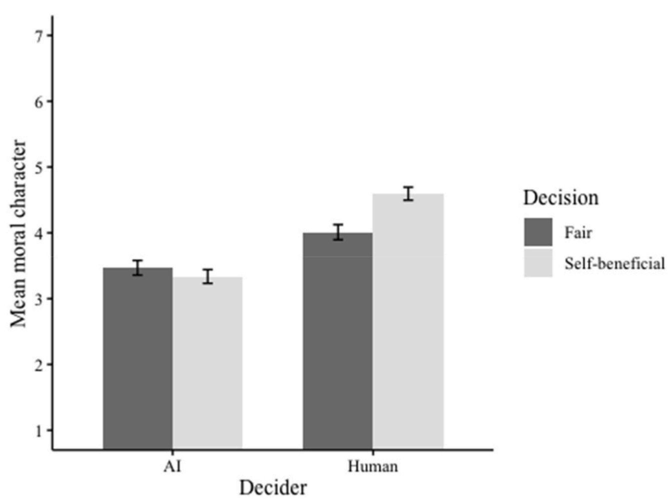

Fig. 9. *Mean moral character judgments as a function of decision and decider (Study 3).*
Note. The error bars represent one standard error.

decision-makers stemmed from liking and mind perception, which was higher for the human (vs. AI) decision-maker. Results of Study 3 show that despite evidence for a strong preference for fairness (Güth et al., 1982), this preference is limited by self-interest, which wins when the two are pinned against each other. As in Study 2, we conducted an exploratory analysis including typing accuracy, mistakes made and words typed as covariates. The results presented remained significant for they key Decider × Self-interest interaction moral character judgements

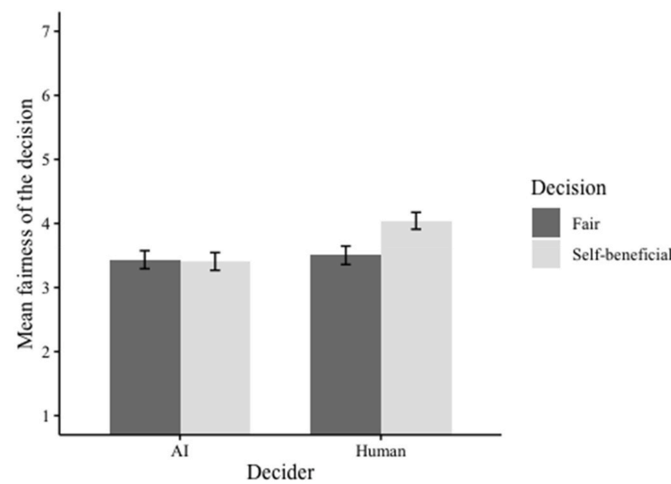

Fig. 10. *Mean decision fairness judgments as a function of decision and decider (Study 3).*
Note. The error bars represent one standard error.

(F(1, 784) = 13.00, $p < .001$, η^2p = .02 [0.005, 0.34]), and fairness (F(1, 784) = 5.31, p = .021, η^2p = .01 [0.001, 0.02]; see Supplementary Materials for full model).

Lastly, this study shows the uniqueness of biased judgments of human decision-makers. For both moral character and fairness variables, the perceptions of AI agents did not differ between fair and disadvantageous and unfair but beneficial conditions, whereas humans gained moral and fairness qualities when making a decision that benefitted an observer relative to fair and disadvantageous conditions. These differences are interesting compared to Study 2, where in both AI and Human self-beneficial conditions the decider was seen as more moral than the other-beneficial decider, but the effect was stronger for the human condition. Compared to the Study 2, we can observe the two motivations – one for fairness and one for self-benefit at stake here, and in the case of the AI condition, the self-benefit did not override the fairness motivations, resulting in no difference between these two conditions in the AI condition. This interesting result indicates that fair AI decisions can partially mitigate biased moral cognition when our interests are at stake. This potential could be explained by the fact that when people's interests are involved in fair decisions, AI agents are liked less and perceived as having a less completed mind than human agents. Consequently, the mechanisms of self-interest bias cannot arise and operate, limiting the impact of self-serving justifications of their moral and fairness judgments. What could supplement this view, is that overall, errors from AI are much less accepted than from humans (Lenskjold et al., 2024) and result in harsher judgments (Jones-Jang & Park, 2022). In effect, AI gains less in terms of moral and fairness

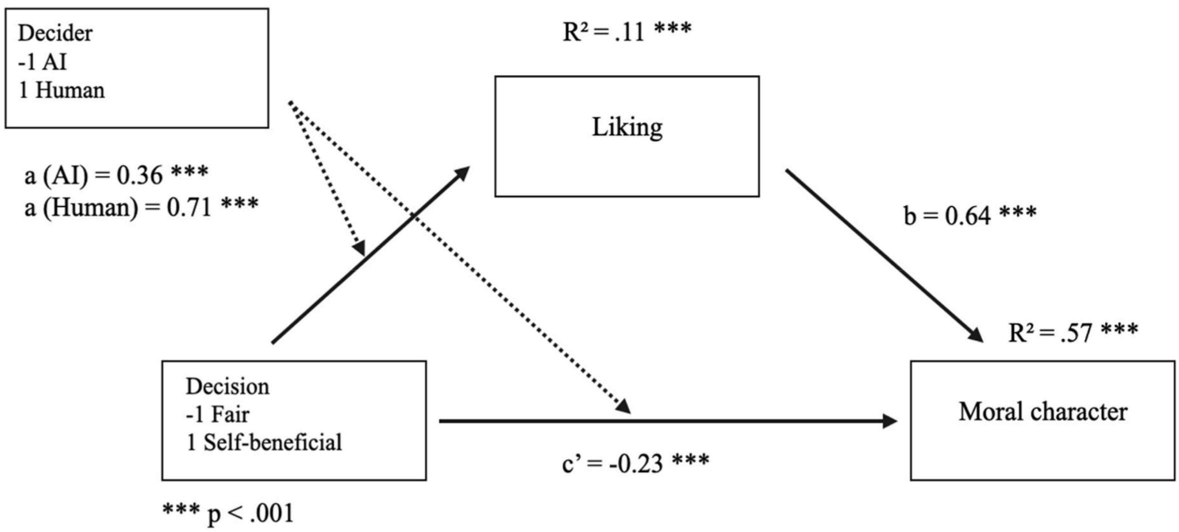

Fig. 11. Model for the moderated mediation of the effect of self-interest on moral character judgments through liking (Study 3).

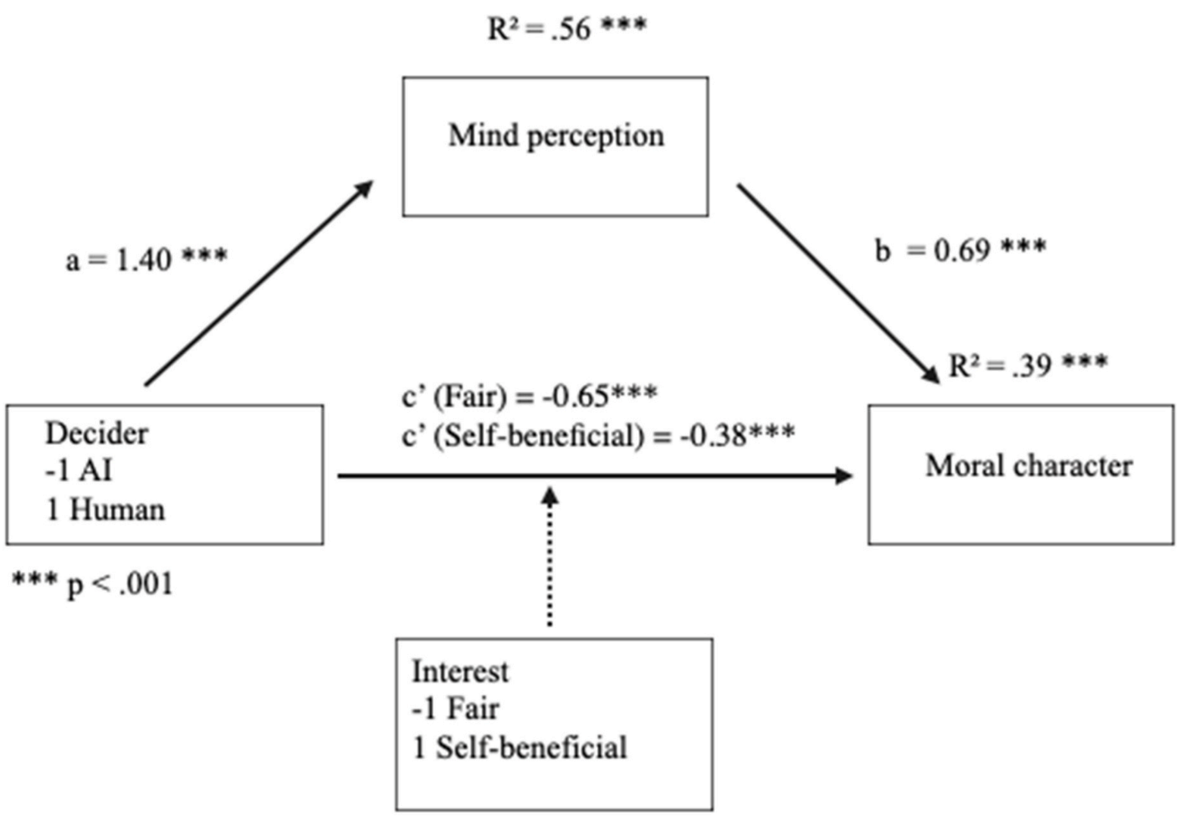

Fig. 12. Model for the moderated mediation of the effect of the agent on moral character judgments through mind perception (Study 3).

judgements from the self-interest bias, because it is also penalized for not behaving as was expected.

13.1. General discussion

In three experiments, we investigated how self-interest, a facet of egocentric perspective, impacts judgments of fairness, moral character, mind perception, and liking of AI and human decision-makers. Across three studies, we show that unfair decision-makers who benefit the observer were judged as more moral and fairer (Studies 1–3). However, these judgments differed when comparing who made the decision. Specifically, the impact of self-interest on moral character judgments was stronger for human decision-makers than for AI decision-makers when the final decision was unfair but beneficial for participants compared to an unfair decision beneficial for another person (Study 2) or fair decision but disadvantageous for participants (Study 3). In Study 3, the self-interest bias was eliminated in moral and fairness judgments of AI decision-makers but persisted in moral perceptions of human decision-makers. Further, we confirmed that those differences between AI and human agents were driven by lower levels of liking and lower perception of mental qualities toward AI than humans (Studies 2 & 3).

13.2. Novel contribution and theoretical significance

While extensive research has documented how self-interest biases fairness and moral judgments (Bocian et al., 2020; Epley & Caruso, 2004), prior work has focused almost exclusively on human agents. Our study extends this literature by examining whether these egocentric biases apply similarly to artificial intelligence (AI) decision-makers. This is critical given AI's growing role in high-stakes decision-making contexts such as hiring, healthcare, and criminal justice (e.g. National Institute of Justice, 2018; Obermeyer et al., 2019).

Furthermore, our work aligns with the Machine Behavior framework (Rahwan et al., 2019), which emphasizes the importance of studying AI as an autonomous agent with social and ethical implications rather than merely as a computational tool. Our findings suggest that fairness and moral judgments are not only biased by self-interest but are also modulated by the perceived mind of the decision-maker (Bigman & Gray, 2018) and attitudinal changes (Bocian et al., 2018). Additionally, our results challenge core assumptions of the Computers Are Social Actors (CASA) paradigm (Nass & Moon, 2000), which posits that people instinctively apply human social norms to AI. If this were entirely the case, the self-interest bias should be equally strong for AI and human decision-makers. However, our findings indicate that AI is judged differently due to its perceived lack of agency and experience (Gray et al., 2012) and differences in liking (Bocian et al., 2018). This suggests that people engage with AI decisions through a distinct cognitive lens that reduces attitudinal and moral engagement relative to human agents.

Overall, our studies not only replicate the well-documented self-interest bias in moral cognition but also extend this work by demonstrating how and why AI decision-makers are evaluated differently from humans. Importantly, we approach this question using preregistered, behavioral experiments involving real monetary consequences, moving beyond the limitations of hypothetical vignette designs. While we acknowledge certain methodological limitations in the following section, using behavioral measures allows us to capture participants' reactions to consequential decisions, not just imagined scenarios. These insights contribute to moral psychology and human-AI interaction research, highlighting the need to consider cognitive biases in designing, implementing, and governance of AI systems.

14. Limitations, implications and future directions

Our line of research has critical implications for algorithmic fairness and public trust in AI, as AI-based decision-making gets adopted more broadly in hiring, legal sentencing or loan approvals. While AI developers focus on minimizing bias in model outputs (e.g., Bellamy et al., 2019; Chen et al., 2023; Holstein et al., 2019, pp. 1–16), our research suggests that public acceptance of AI fairness may be equally shaped by human biases, particularly egocentric distortions in moral reasoning. If people judge fairness based on self-interest rather than objective principles, AI fairness interventions may need to account for psychological biases rather than just technical improvements. Future work should explore whether transparency about AI decision-making processes can mitigate egocentric biases, or whether increased anthropomorphism could paradoxically exacerbate them by making AI seem more human-like and, thus, more prone to bias.

For example, Workday, a company providing hiring screening software, has been accused of discrimination in a class-action lawsuit. AI used in their software was allegedly discriminating based on race, age, and disability (Wiessner, 2024). We see action taken to make the algorithms fairer by those who lost access to resources, not those who got the job despite the unfairness. Our studies shed light on this issue, showing that those who benefit from the unfairness do not see it as unfair and immoral, and this issue equally regards those who are recipients of algorithm decisions and those who design and implement them.

Although it has not been tested directly in this line of studies, evidence suggests that people are unaware of self-interest bias (Bocian & Wojciszke, 2014b). Therefore, there is a high probability that when people benefit from algorithm decisions, they would also be less likely to act upon the unfairness and take action to make the situation more equal for everyone. Future studies could further investigate whether people are willing to act against unfairness and injustice in AI decisions that benefit them.

The strength of this research line lies in the use of behavioral methods in manipulating self-interest (Studies 1–3). Hence, participants could experience the financial consequences of AI's decisions, not just imagine them. As highlighted in a recent review (Mahmud et al., 2022), most studies on algorithmic decision-making rely on vignette-based designs, which come with important limitations. More broadly, there is a growing call to incorporate behavioral methods in AI research (Mahmud et al., 2022) and in psychological science (Doliński, 2018). The methodological choices made in prior work may help explain some discrepancies between our findings and those reported in the existing literature.

For example, Wilson et al. (2022) found that AI was seen as less wrong than humans for the same immoral actions. Additionally, Bigman et al. (2023) found that AI causes less moral outrage for discrimination than humans. Alternatively, Bankins et al. (2022) found that when a decision had a positive outcome for the observers, the human was judged as more just than AI. In our studies, AI was ascribed fewer moral traits than humans, making the same unfair decision (Studies 2 & 3). As most studies about the perceptions of AI use vignettes (Mahmud et al., 2022), including the studies mentioned above, people may perceive imagined scenarios differently than when they take part in them. Alternatively, this may be explained by the direct involvement of participants' interests, not only the overall valence of the situation. Future studies could explore the differences between imagined scenarios vs. behavioral designs on how people perceive the decision-making agents.

Nevertheless, we acknowledge that our work has limitations that might warrant future research. One limitation of our study is its relatively abstract decision-making context with simulated AI interaction, which may limit ecological validity. While methodologically clean designs allow for precise identification of effects, future research should examine alternative approaches that minimize deception and incorporate more realistic use-cases. For instance, using actual AI-driven decision-making systems, such as automated hiring, credit scoring, or healthcare diagnostics. This would allow us to move beyond theory to real-world uses of AI and provide more ecologically valid insights.

Secondly, the current task was relatively low-stakes, resulting in a minimal financial gain (up to 0.75 bonus). Our choice of task was based on previous research showing that people believe they are better than average in simple tasks (Kruger, 1999) and that writing on a keyboard is a common ability (Pinet et al., 2022). Future research should investigate whether self-interest biases AI fairness perceptions differently in high-stakes environments, such as workplace hiring decisions or criminal justice settings.

Additionally, while our studies were conducted with U.S. participants, future work should explore cross-cultural differences in AI moral perceptions—particularly in societies with differing levels of fears towards AI (Dong et al., 2024). For instance, recent studies have shown that cooperation with AI can differ significantly across countries (Karpus et al., 2025), highlighting the importance of testing whether the strength of the egocentric bias we observed generalizes beyond Western contexts. Given that AI decision-making is a global issue, understanding whether different cultures exhibit different sensitivities to such biases is essential for both theory and the ethical implementation of AI systems.

Our findings confirm that self-interest bias is mediated by liking—decision-makers who made self-beneficial unfair decisions were liked more, which in turn led to more favorable moral character judgments. However, this effect was significantly weaker for AI than for human decision-makers, suggesting that affective responses play a reduced role in moral evaluations of AI. While it is important to recognize that liking does not necessarily “cause” changes in moral character judgments, it likely reflects an affective bias that amplifies self-interest effects. Past research supports this interpretation, indicating that liking is a key psychological mechanism in the self-interest bias (Bocian et al., 2018; Bocian & Wojciszke, 2014a).

First, studies using a moderation-of-process design (Spencer et al., 2005) have shown that when liking is experimentally blocked, self-interest bias does not emerge (Bocian & Wojciszke, 2014a). This suggests that liking is not merely a correlational byproduct but a functional component of self-interest effects. Second, direct manipulations of liking lead to inflated moral character judgments, even when liking is induced through unrelated factors that are irrelevant to morality (Bocian et al., 2018). Together, these findings highlight the central role of affective responses in moral judgment formation and suggest that AI decision-makers may be evaluated differently because they do not elicit the same social and emotional engagement as human agents.

Additionally, our findings on mind perception as a mediator require further investigation. While prior research suggests that higher perceived agency leads to stronger moral judgments (Bigman & Gray, 2018), our results indicate that self-interest biases can distort mind perception. Specifically, participants attributed more agency and experience to decision-makers whose choices benefited them. This suggests a novel cognitive distortion in which people project greater “mindedness” onto agents who serve their self-interest.

Given recent critiques of mediation analyses (Rohrer et al., 2022), we acknowledge that alternative causal explanations may exist. Future research should experimentally manipulate liking and mind perception to test causal relationships more directly. For example, artificially enhancing an AI’s human-like traits (e.g., by adding facial expressions or emotional responses) could reveal whether mind perception drives moral attributions or merely correlates with them. Past research shows that manipulating certain traits, such as the gender of a bot, increases perceptions of humanness (Borau et al., 2021), and that manipulating features like name, gender, and voice increases anthropomorphization, which in turn boosts trust in autonomous vehicles (Waytz et al., 2014). Additionally, explicitly manipulating liking through experimental interventions (e.g., inducing positive vs. negative attitudes toward decision-makers) would help determine its precise role in moral evaluations.

15. Implications for AI fairness and public trust

Our findings have direct implications for AI ethics, policy, and design. Given that people judge human decision-makers more favorably than AI for self-beneficial decisions, AI systems may need additional transparency mechanisms to be perceived as fair. Organizations deploying AI-driven hiring, loan approval, or healthcare recommendations should consider whether AI’s lack of perceived moral agency might influence user acceptance and compliance with decisions (Hoff & Bashir, 2015).

Our results suggest that people may be less biased in their moral evaluations when AI makes fair decisions because AI lacks the affective pull that human decision-makers exert. This raises the intriguing possibility that fair AI systems might reduce self-interest biases in decision-making. One potential intervention could be designing AI systems that explicitly highlight procedural fairness rather than focusing on human-like qualities that could increase likability biases.

These findings underscore the need for AI literacy programs that educate the public on algorithmic fairness and bias. Since individuals apply self-interest biases even to AI agents, policymakers should consider how to frame AI decisions to mitigate egocentric distortions. For instance, presenting AI decisions alongside explanations of fairness criteria may help individuals evaluate decisions more objectively.

As AI continues to play a larger role in societal decision-making; understanding how human biases shape moral judgments of AI decisions remains critical. Future research should explore whether increasing AI’s perceived agency and intentionality would make AI decisions more or less susceptible to human biases impacting their moral judgments, particularly in high-stakes contexts such as law enforcement and healthcare.

16. Conclusion

This paper examines how self-interest influences moral character and fairness judgments of decision-makers, whether human or AI. When individuals personally benefit from an unfair decision, they are less likely to perceive it as immoral or unjust, regardless of whether the decision-maker is human or artificial. However, people tend to evaluate AI more critically because AI is generally perceived as having lower mental capacities and less moral agency than humans (Bigman & Gray, 2018; Gray et al., 2012). As a result, the self-interest bias in moral evaluations is weaker for AI than for human decision-makers. This suggests that while people justify unfair actions when they serve their own interests, they are less inclined to extend such leniency to AI-driven decisions, holding AI to a stricter ethical standard.

Data archiving and accessibility statement

The data are openly available at https://osf.io/qpkaj/.

CRediT authorship contribution statement

Katarzyna Miazek: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Konrad Bocian:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization, Methodology.

Ethical statement

The reported studies were approved by the ethical committee of the SWPS University (Ethics Clearance ID: WKE/S2023/06/07/129). All participants provided informed consent.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chbr.2025.100719.

Data availability

Data are available at https://osf.io/qpkaj/.

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